

Mixed Aliphatic Organozinc Reagents as Nonstabilized C_{sp}³-Nucleophiles in the Multicomponent Mannich Reaction

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Cite This: *J. Org. Chem.* 2022, 87, 4961–4964



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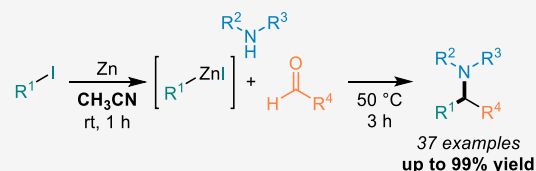


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ABSTRACT: The use of mixed aliphatic organozinc reagents in the multicomponent Mannich reaction is described. A large variety of primary, secondary, and tertiary organozinc reagents, secondary amines and aromatic or aliphatic aldehydes could be used for the straightforward preparation of densely substituted amines. The three-component reaction could additionally be performed starting from alkyl halides under reductive (Barbier-type) conditions.



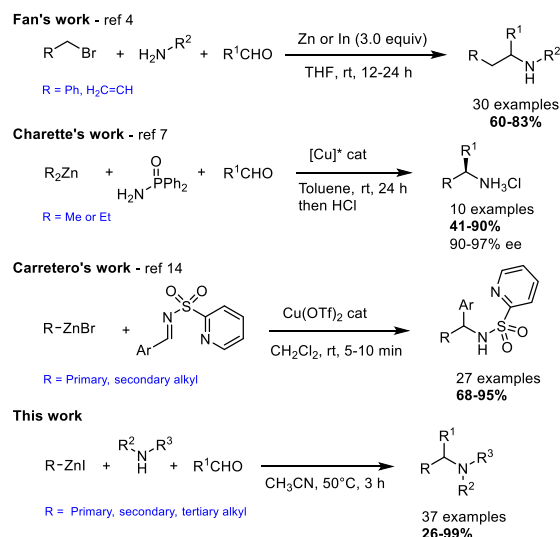
Mild conditions

No additional metals

Important scope

The Mannich reaction is a classic example of multicomponent reaction and represents one of the most powerful tool of organic synthesis.¹ Indeed, it allows the straightforward preparation of α -branched amines with the formation of both a C–C bond and a C–N bond in a single step.² Among recent developments, the use of organometallic species as nucleophiles (organometallic multicomponent Mannich reaction) has greatly expanded the scope of the original transformation.³ Therefore, significant contributions have been implemented using sp-, sp²-, or sp³-hybridized organometallic derivatives, but this last category is restricted to allyl⁴ or benzyl^{4,5} organometal species. In this context, examples of nonstabilized sp³-hybridized organozinc reagents as one reaction partner are more limited and mainly involve dialkylzinc reagents, due particularly to their substantial reactivity and functional group tolerance.^{6,7} However, the scope of the corresponding Mannich reactions is hampered by obstacles inherent to their preparation, and mainly commercially available or easily accessible dialkylzinc species (Me₂Zn, Et₂Zn) have been employed so far. In contrast, mixed alkylzinc species (RZnX) are more easily accessible and usually prepared by direct insertion of zinc metal into alkyl halides.⁸ Thus, Knochel described the insertion of zinc in alkyl iodides in THF at 30 °C,⁹ whereas Rieke showed the insertion of activated zinc in alkyl bromides in THF at rt.¹⁰ More recently, Knochel introduced the use of Zn powder in the presence of LiCl in THF for the derivatization of alkyl bromide at rt,¹¹ and Huo described the insertion of zinc dust in alkyl bromides at 80 °C in DMA or DMF.¹² They can additionally be prepared by transmetalation strategies.¹³ However, despite the number of methods allowing their straightforward preparation, their use as one reaction partner in multicomponent Mannich reactions remains elusive. Indeed, mixed alkylzinc species, preformed prior to use, were only employed by Carretero in related nucleophilic additions on activated imines under Cu catalysis (Scheme 1).¹⁴

Scheme 1. Selected Examples of Organometallic Mannich Reactions Involving Csp³-Organozinc Reagents



Therefore, the development of a truly multicomponent Mannich reaction employing mixed alkylzinc reagents would constitute a relevant alternative to classical methods^{15,16} for the straightforward preparation of α -branched amines.

We thus went in search for new conditions allowing the synthesis of mixed alkylzinc reagents and their consecutive use in multicomponent Mannich reactions. Our experience in the

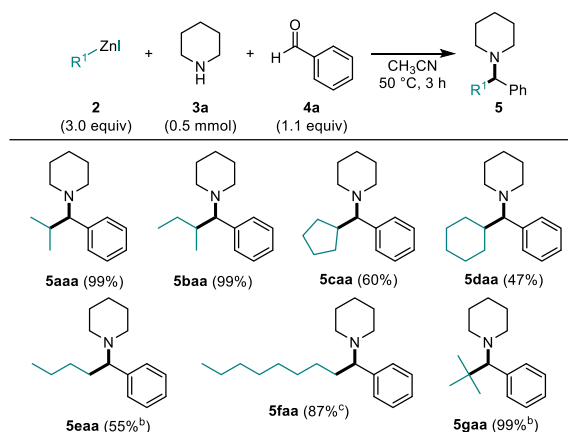
Received: December 9, 2021

Published: March 10, 2022



development of multicomponent reactions prompted us to consider the use of acetonitrile as a key element of the process.^{17,18} This assumption could be verified again. Indeed, preliminary experiments indicated that this solvent was the most efficient for both the zincation step and the subsequent three-component reaction, which was nearly quantitative after 3 h at 50 °C. Due to these encouraging results, it was decided to examine the reaction between piperidine (**3a**, 1.0 mmol), benzaldehyde (**4a**, 1.1 equiv), and an excess (3.0 equiv) of an organozinc reagent **2**, preformed in acetonitrile from alkyl iodides **1** in yields up to 90% after only 30 min at room temperature (Scheme 2).¹⁹

Scheme 2. Scope of Organozinc Reagents^a

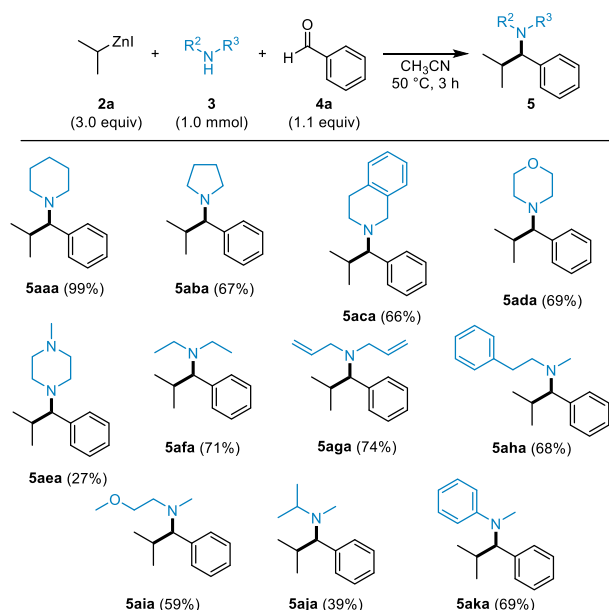


^aYields of isolated products. Reaction conditions: **3a** (0.5 mmol), **4a** (1.1 equiv), **2** (3.0 equiv), 50 °C, 3 h. ^bReaction performed with 1.0 equiv of TMSCl. ^cReaction performed with 1.0 equiv of TMSCl at 80 °C for 14 h.

Under these conditions, acyclic secondary iodides were very efficient (**5aaa** and **5baa**, 99%) in the process, but the use of cyclic starting materials **2c** and **2d** led to reduced yields of 60% (**5caa**) and 47% (**5daa**), respectively. In the case of organozinc reagents obtained from primary iodoalkanes, Mannich reactions were much less efficient than expected and required extended reaction time and increased amounts of TMSCl to furnish products **5eaa** and **5faa** in satisfactory to good yields (55 and 87% respectively). Interestingly, the use of the more hindered tertiary organozinc reagent **2g** was also possible and even efficient under the same reaction conditions (**5gaa**, 99%). Unfortunately, all attempts to use functionalized aliphatic iodides were unsuccessful because of the random reactivity of the corresponding organozinc reagents.

We next studied the reaction scope of the amine partner (Scheme 3). Thus, the organozinc reagent **2a** deriving from 2-iodopropane was allowed to react with benzaldehyde (**4a**) and a variety of secondary amines **3** (primary amines proving inoperant in the present reaction conditions). The use of cyclic secondary amines led to similar good yields of 66–69%, except for *N*-methylpiperazine, which led to only 27% of the expected product **5aea**. The use of acyclic amines such as diethylamine (**3f**) or diallylamine (**3g**) led to slightly higher yields of 71% and 74%, respectively. Other *N*-methylamines, including *N*-methylaniline (**3k**), could be used with similar efficiencies, but a lower yield was obtained in the case of *N*-isopropylmethylamine (**3j**). These results confirm the importance of steric hindrance over the amine nucleophilicity in acetonitrile.^{20,21}

Scheme 3. Scope of Secondary Amines^a

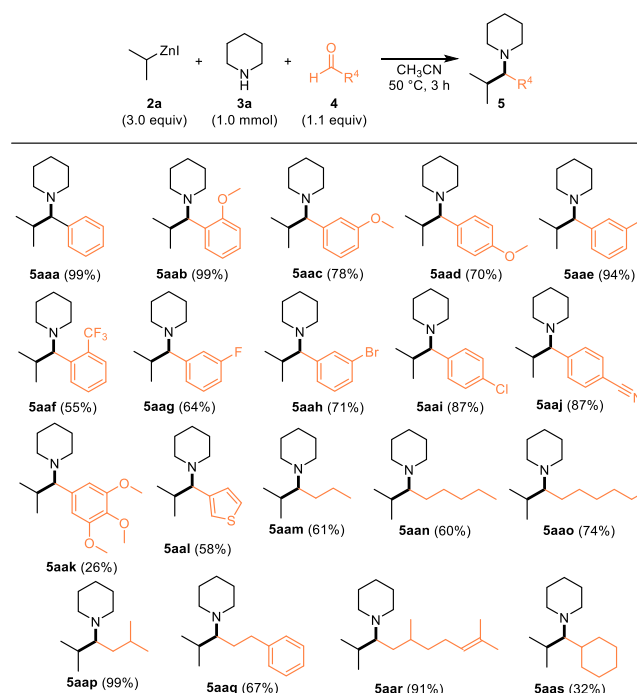


^aYields of isolated products. Reaction conditions: **3** (1.0 mmol), **4a** (1.1 equiv), **2a** (3.0 equiv), 50 °C, 3 h.

The present Mannich reactions also offered an important diversity regarding the aldehyde partner (Scheme 4).

A broad range of aromatic aldehydes could be used in good to excellent yields. As illustrated in the case of a methoxy group, substitution was possible at any position (*ortho*, *meta*, *para*) of the aromatic ring. Diverse functional groups could be used, including a methyl group (**5aae**, 94%), a trifluoromethyl group (**5aaf**, 55%), a halogen atom (F: **5aag**, 64%; Br: **5aah**,

Scheme 4. Scope of Aldehydes^a

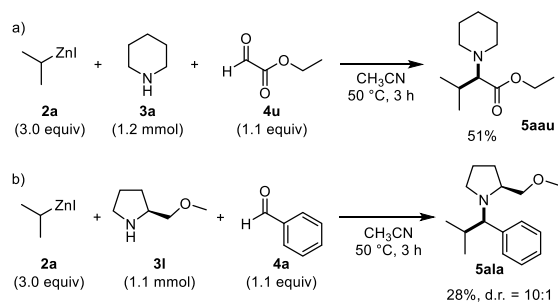


^aYields of isolated products. Reaction conditions: **3a** (1.0 mmol), **4** (1.1 equiv), **2a** (3.0 equiv), 50 °C, 3 h.

71%; Cl: **5aai**, 87%), or a cyano group (**5aaj**, 87%). From a general point of view, electron-withdrawing groups afforded better yields than electron-donating groups. However, in the case of *ortho*-substituents, whereas steric hindrance seemed predominant in the case of the trifluoromethyl group (**5aaf**, 54%), this was not the case for a chelating group like methoxy (**5aab**, 99%), probably indicating favorable secondary interactions at the transition state (see below). Densely functionalized benzaldehydes derivatives or a heteroaromatic aldehyde could also be used, albeit with lower yields (**5aak**, 26% and **5aal**, 58%). For **5aak**, the lack of reactivity may be attributed to the moderate electrophilicity of the starting aldehyde. Importantly, aliphatic aldehydes proved perfectly suitable under the present reaction conditions. Therefore, linear aldehydes such as butyraldehyde, hexanal or heptanal afforded the expected products **5aam**, **5aan**, and **5aao** in similarly good yields (60–74%), and ramification (**5aap**, 99%) or substitution (**5aaq**, 67%) of the alkyl chain was also possible. More complex aldehydes such as ($\pm$)-citronellal could be used efficiently (**5aar**, 91%). A more hindered aliphatic aldehyde also underwent the reaction as illustrated by the successful use of cyclohexane carboxaldehyde (**5aas**, 99%). However, pivalaldehyde proved unreactive under these reaction conditions.

Two additional examples were found particularly interesting (Scheme 5). Thus, the Mannich reaction performed between

Scheme 5. Useful Multicomponent Couplings

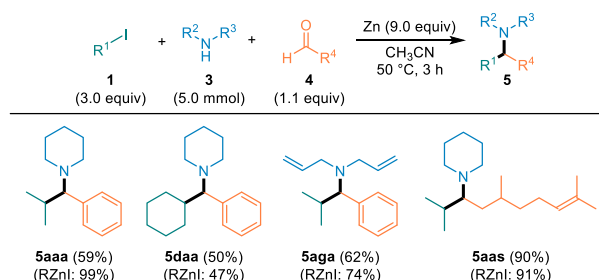


2a, **3a**, and ethyl glyoxylate (**4u**) afforded **5aau** in 51% yield. This reaction constitutes an original and straightforward access to racemic unnatural valine derivatives. On the other hand, 1-pyrrolinol derivative **3l** proved to be a promising substrate for the stereocontrol of the reaction as it led to a good asymmetric induction (10:1 dr) on the newly formed stereocenter.

To fully demonstrate the usefulness of the present methodology, we finally investigated the feasibility of such Mannich reactions under Barbier (reductive) conditions,²² hence generating the organozinc species in the presence of the other reaction partners. Indeed, this modification would lead to an original reductive multicomponent coupling reaction, thus increasing the experimental simplicity of the present transformation. It can be noted that besides works involving allylic or benzylic halides, such reductive multicomponent couplings have also been previously demonstrated for sp^2 -hybridized halides under transition-metal catalysis.^{23,24} To this extent, a suspension of activated zinc dust in acetonitrile was successively treated with 2-iodopropane (**1a**), benzaldehyde (**4a**), and piperidine (**3a**), and the reaction was carried out under standard conditions (50 °C for 3 h). After treatment of the reaction mixture, we were pleased to isolate **5aaa** in an encouraging 59% yield. We thus performed a quick survey of

the reaction system under these new conditions, which revealed that the expected Mannich adducts can be obtained in yields generally similar to those observed with preformed organozinc reagents (Scheme 6). It can be noted that, surprisingly, no reaction occurred with tertiary iodides under such reductive conditions.

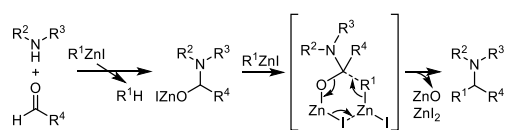
Scheme 6. Mannich Reactions under Reductive Conditions⁴⁴



^aYields of isolated products. Reaction conditions: **1** (3.0 equiv), **3** (5.0 mmol), **4** (1.1 equiv), Zn (9.0 equiv) activated with TFA and 1,2-dibromoethane, CH_3CN ($C = 0.5$ M), 50 °C, 3 h.

In sharp contrast with other works describing the preparation of related products from alkyl iodides via a radical mechanism in the presence of radical initiators,²⁵ the present transformation is presumed to occur via a polar mechanism, similarly to those described in the case of analogous organozinc species (Scheme 7).^{4,26} Therefore, the reaction

Scheme 7. Putative Reaction Mechanism



presumably involves the generation of a hemiaminal by addition of the secondary amine onto the aldehyde followed by deprotonation by a first equivalent of the organozinc reagent. Then, the addition of a second equivalent of the reagent onto the resulting zinc alcoholate would lead to the final product via a putative 6-membered transition state.

In conclusion, we have introduced the use of nonstabilized mixed aliphatic organozinc reagents in the multicomponent Mannich reaction. The reaction occurs under very simple experimental conditions through a polar mechanism. It allows the straightforward preparation of densely substituted amines and displays a broad scope of substrates, making it a presumably relevant alternative to existing methods. An alternative reductive procedure could be successfully employed instead of using preformed organozinc species.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.joc.1c02996>.

Experimental procedures, compound characterization data, and NMR spectra for all compounds (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

Financial support of this work by the CNRS, the University Paris-Est Créteil, and the University Paris-Est (Ph.D. grant to M. Pi) is gratefully acknowledged.

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